ClimateWins Predictions

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Data Analyst

June 21th, 2025

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Objectives and Hypothesis

- **<u>Objective</u>**: Use machine learning to help predict the consequences of climate change.
- **<u>Hypothesis 1</u>**: The climate hasn't changed much from 1973 until 2022.
- **<u>Hypothesis 2</u>**: There was nicer weather about 50 years ago.
- **<u>Hypothesis 3</u>**: 70% of days in Europe aren't days with pleasant weather.

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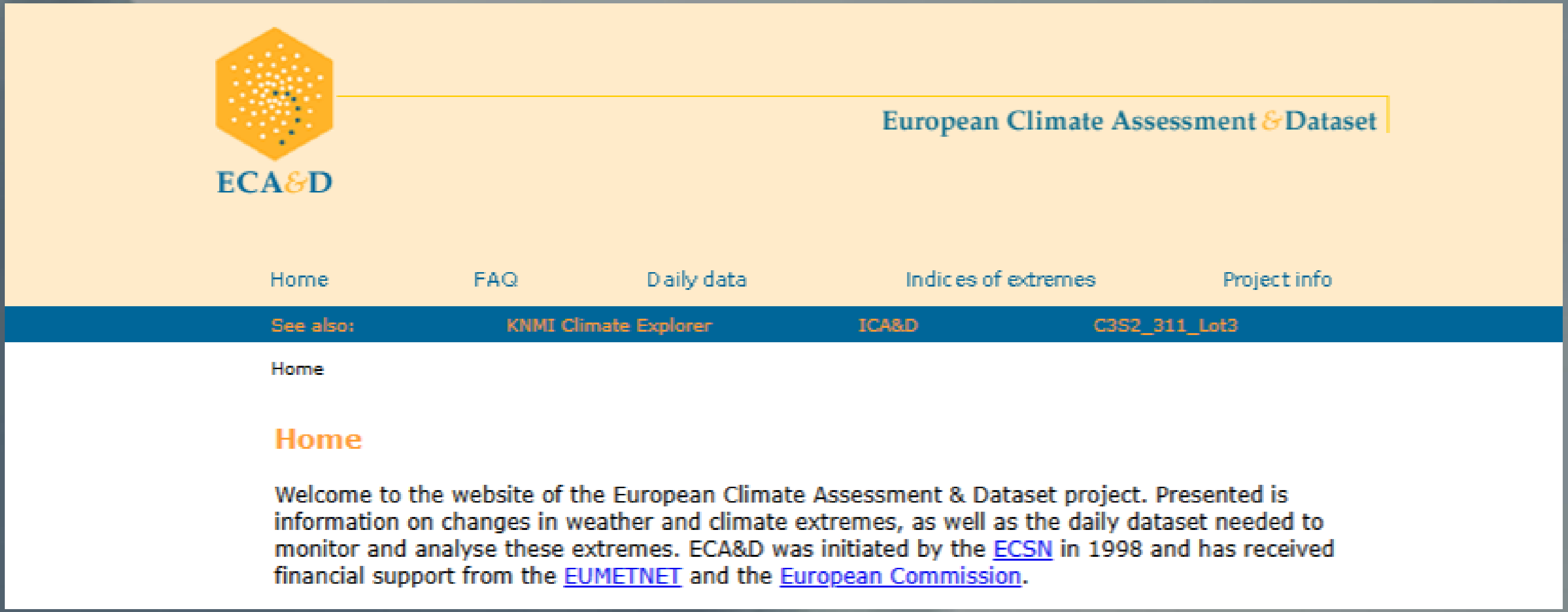
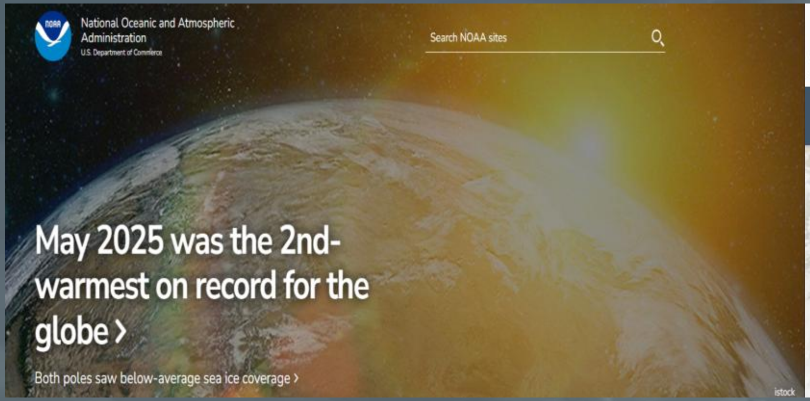
Sources, Biases, Accuracy,

- Data Sources:
  - European Climate Assessment & Data Set project
    - The National Oceanic and Atmospheric Administration (NOAA)
    - The Japan Meteorological Agency (JMA)
- Bias: ML algorithms or personal bias.
- Accuracy: Always a potential for inaccurate results





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Data Optimization

- Gradient Descent was used to optimize the data.
  - **Scaled**
  - Box and Whisker Plot
  - Scatterplot
  - Loss Function with 3D Models

jupyter ML1 - 1.2 Ethics and Directions in MLP Last Checkpoint: 28 days ago

File Edit View Run Kernel Settings Help Trusted

Markdown JupyterLab Python [conda env:base] *

[23]:
```
# Create scaled data\n",
W_scaled = pd.DataFrame(scaler.fit_transform(W), columns=W.columns, index=W.index)
W_scaled.head()
```

[23]:

	DATE	MONTH	BASEL_cloud_cover	BASEL_wind_speed	BASEL_humidity	BASEL_pressure	BASEL_global_radiation	BASEL_precipitation	BASEL_snow_depth	BASEL
0	-1.707663	-1.599964	0.660514	-0.02793	0.826097	-0.001949	-1.101066	-0.265148	-0.179228	
1	-1.707657	-1.599964	0.244897	-0.02793	0.735760	-0.001949	-1.058108	1.658760	-0.179228	
2	-1.707652	-1.599964	1.076130	-0.02793	1.277781	-0.001949	-1.251420	0.155707	-0.179228	
3	-1.707646	-1.599964	-1.001953	-0.02793	1.458455	-0.001949	-0.821838	-0.445514	-0.179228	
4	-1.707641	-1.599964	0.244897	-0.02793	1.729466	-0.001949	-0.746661	-0.164944	-0.179228	

[24]:
```
# Compare Original vs Scaled data\n

W.head()
```

[24]:

	DATE	MONTH	BASEL_cloud_cover	BASEL_wind_speed	BASEL_humidity	BASEL_pressure	BASEL_global_radiation	BASEL_precipitation	BASEL_snow_depth	BASEL_
0	19600101	1	7	2.1	0.85	1.018	0.32	0.09	0	
1	19600102	1	6	2.1	0.84	1.018	0.36	1.05	0	
2	19600103	1	8	2.1	0.90	1.018	0.18	0.30	0	
3	19600104	1	3	2.1	0.92	1.018	0.58	0.00	0	
4	19600105	1	6	2.1	0.95	1.018	0.65	0.14	0	

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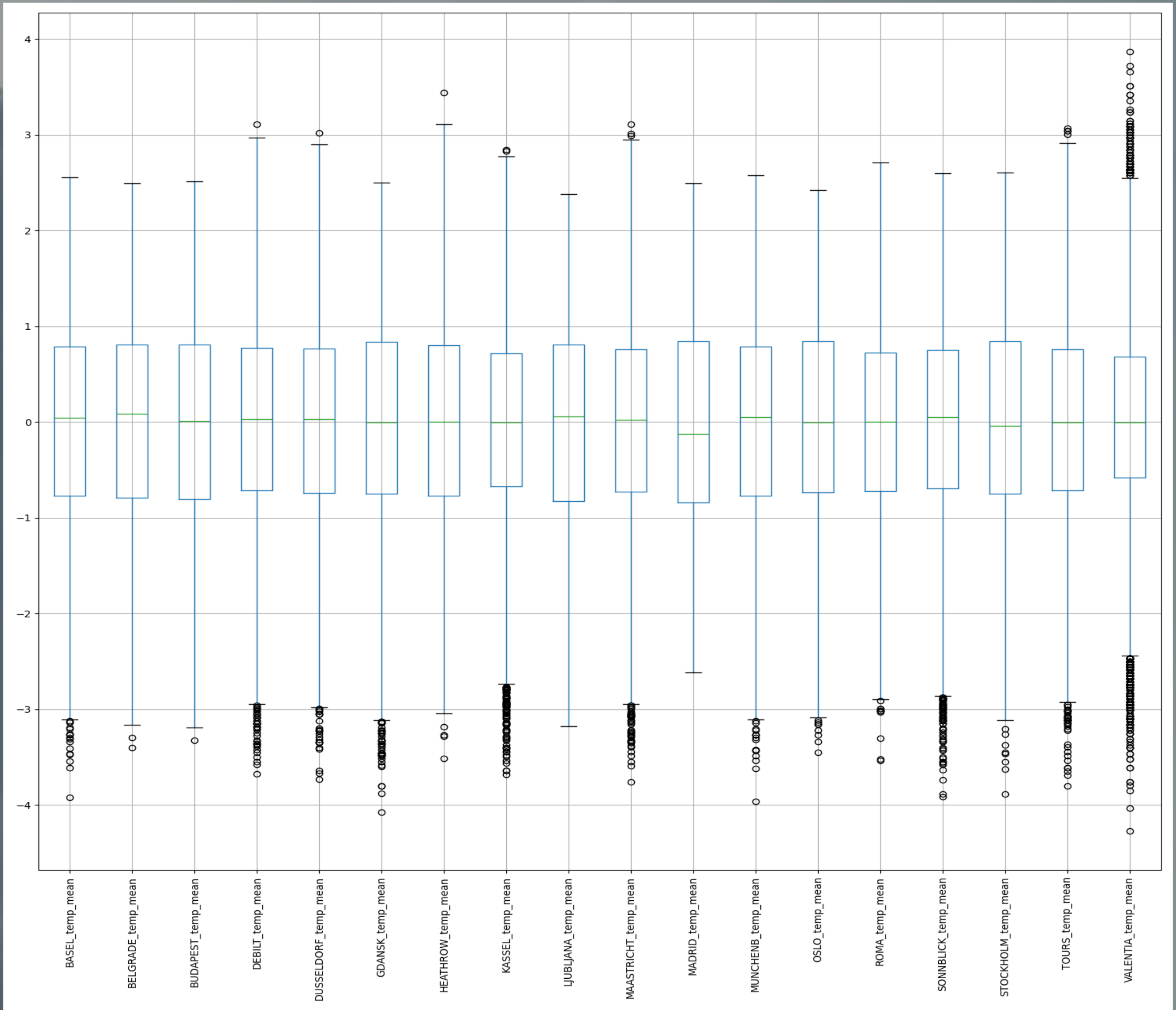

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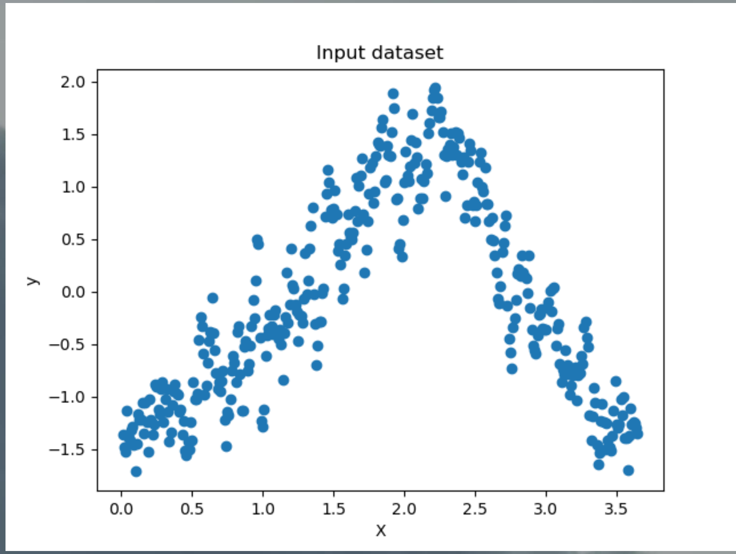
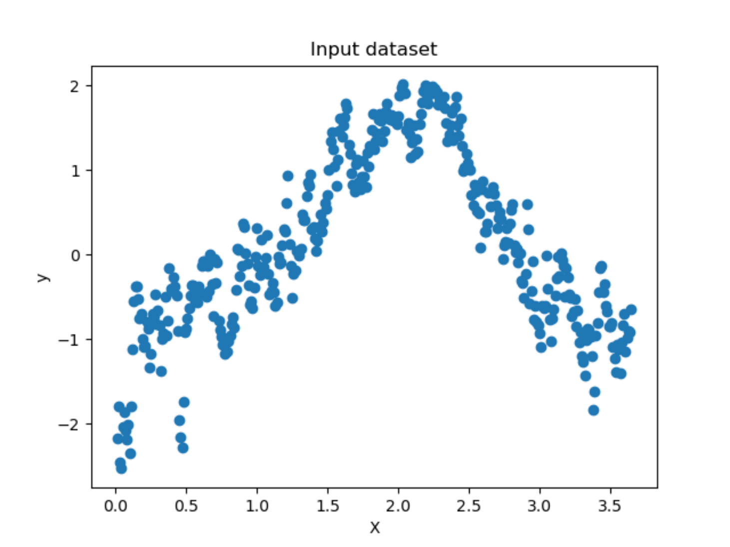
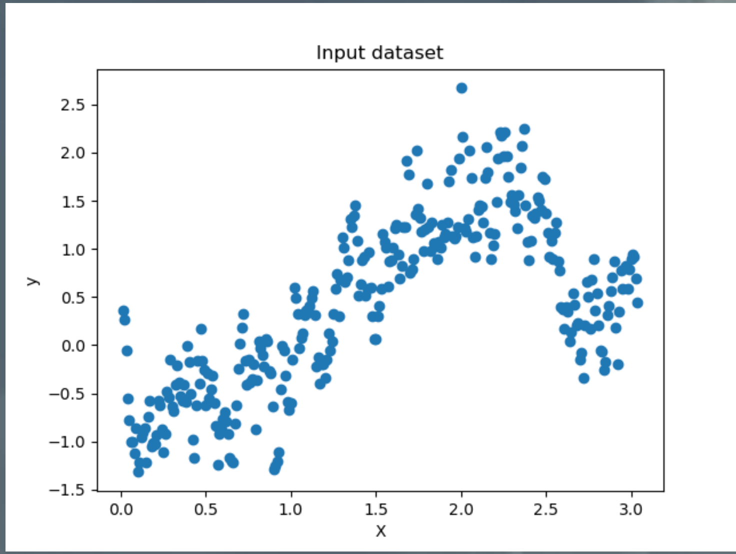


Madrid year 1973



Oslo year 1997



Debilt year 2022

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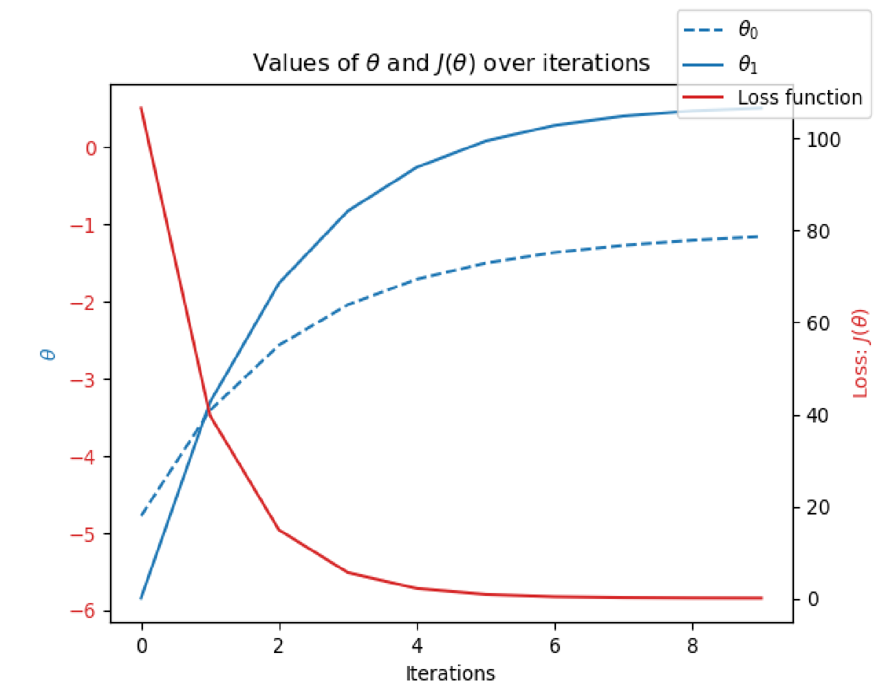

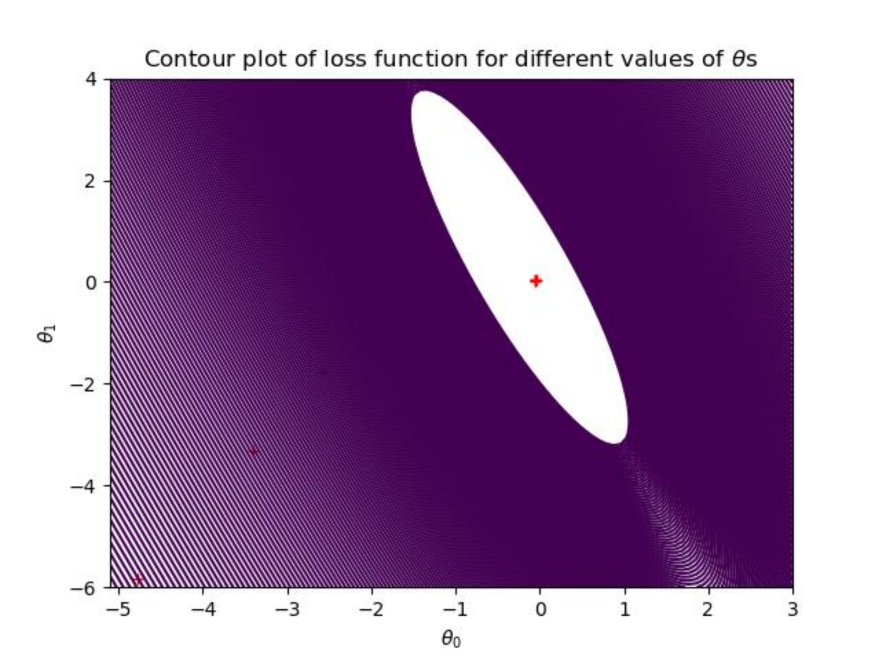

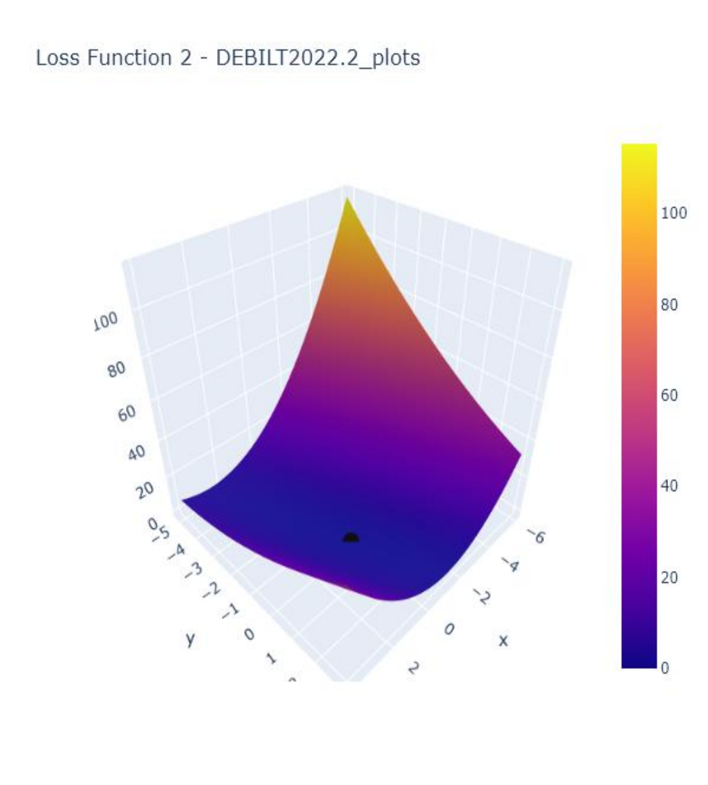

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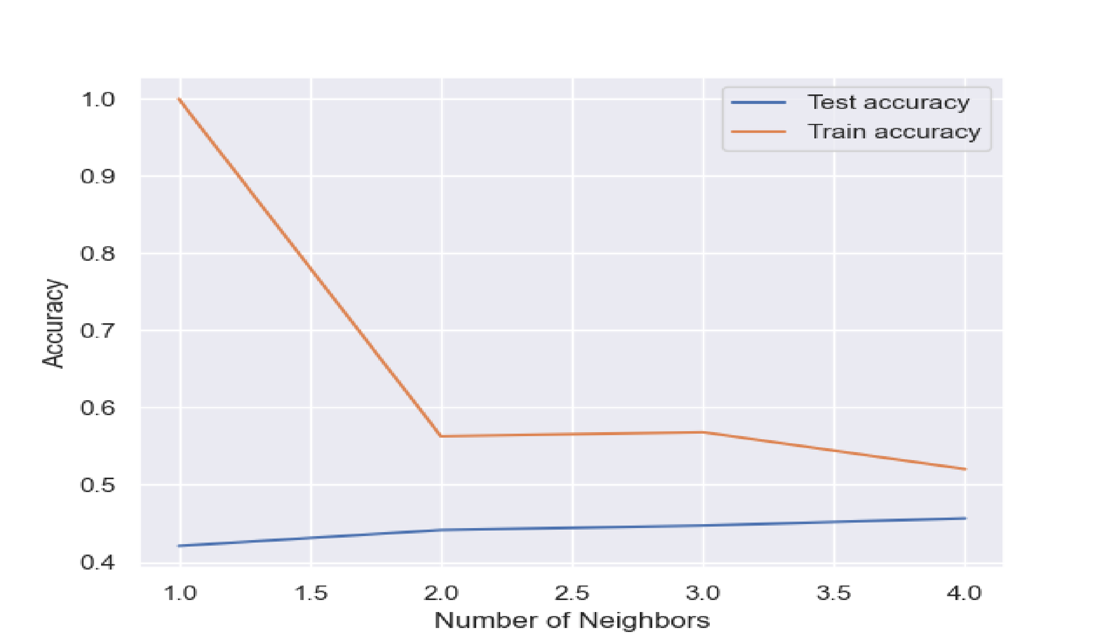
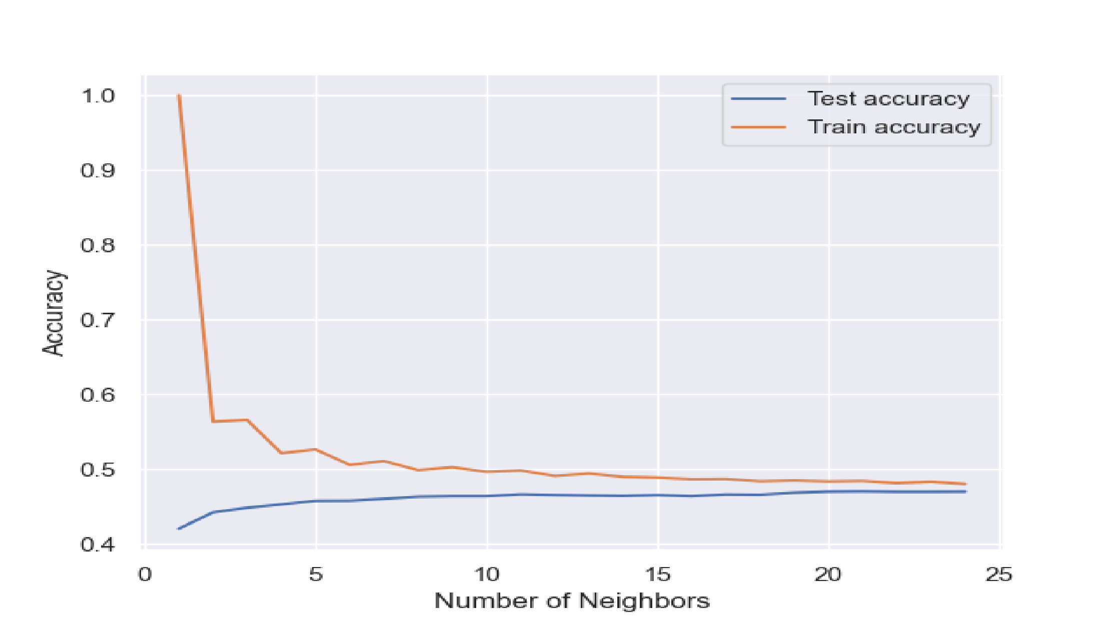
Supervised Learning Algorithms Best Fit

- **K-Nearest Neighbor**
- Decision Tree
- Artificial Neural Network





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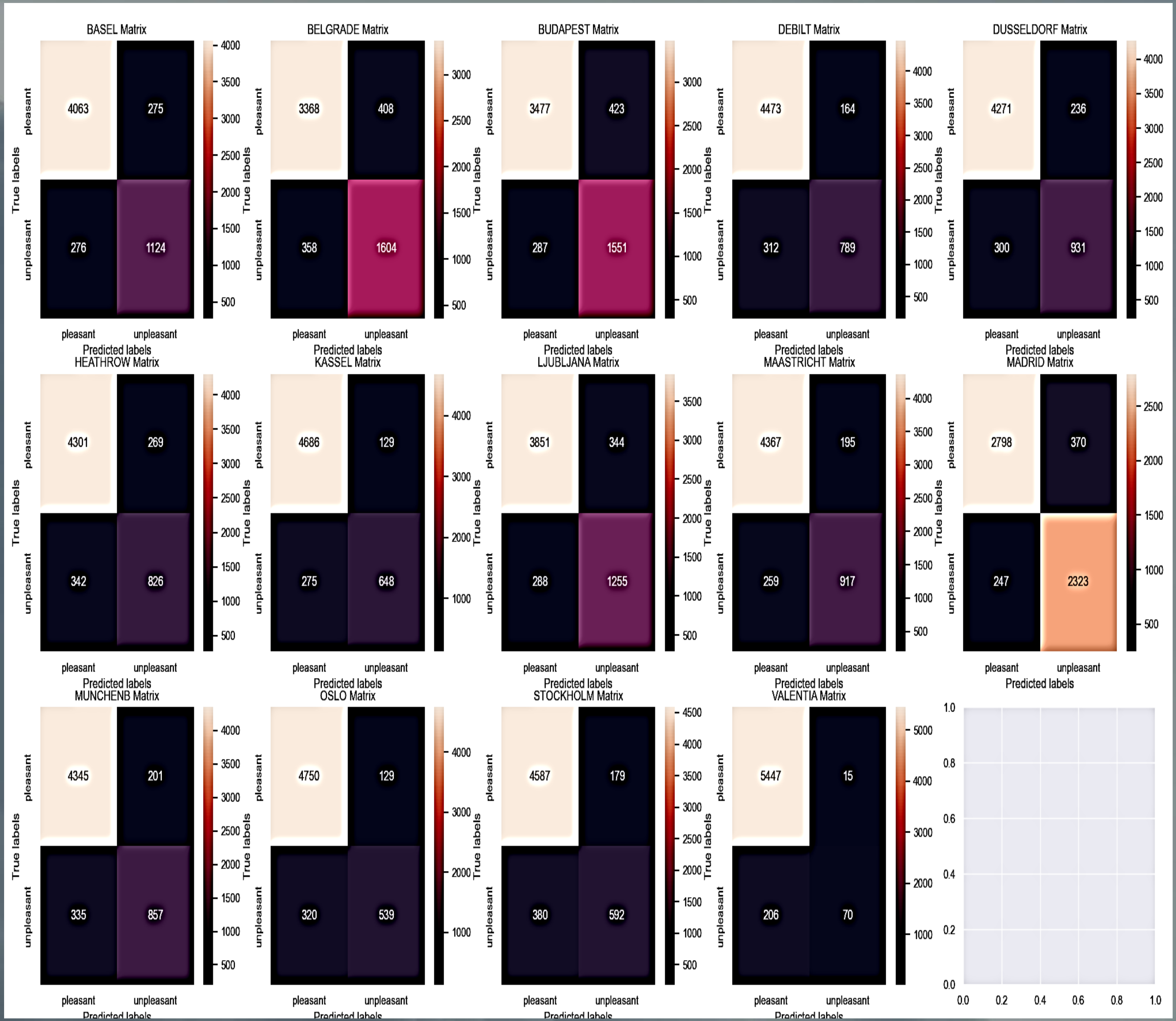

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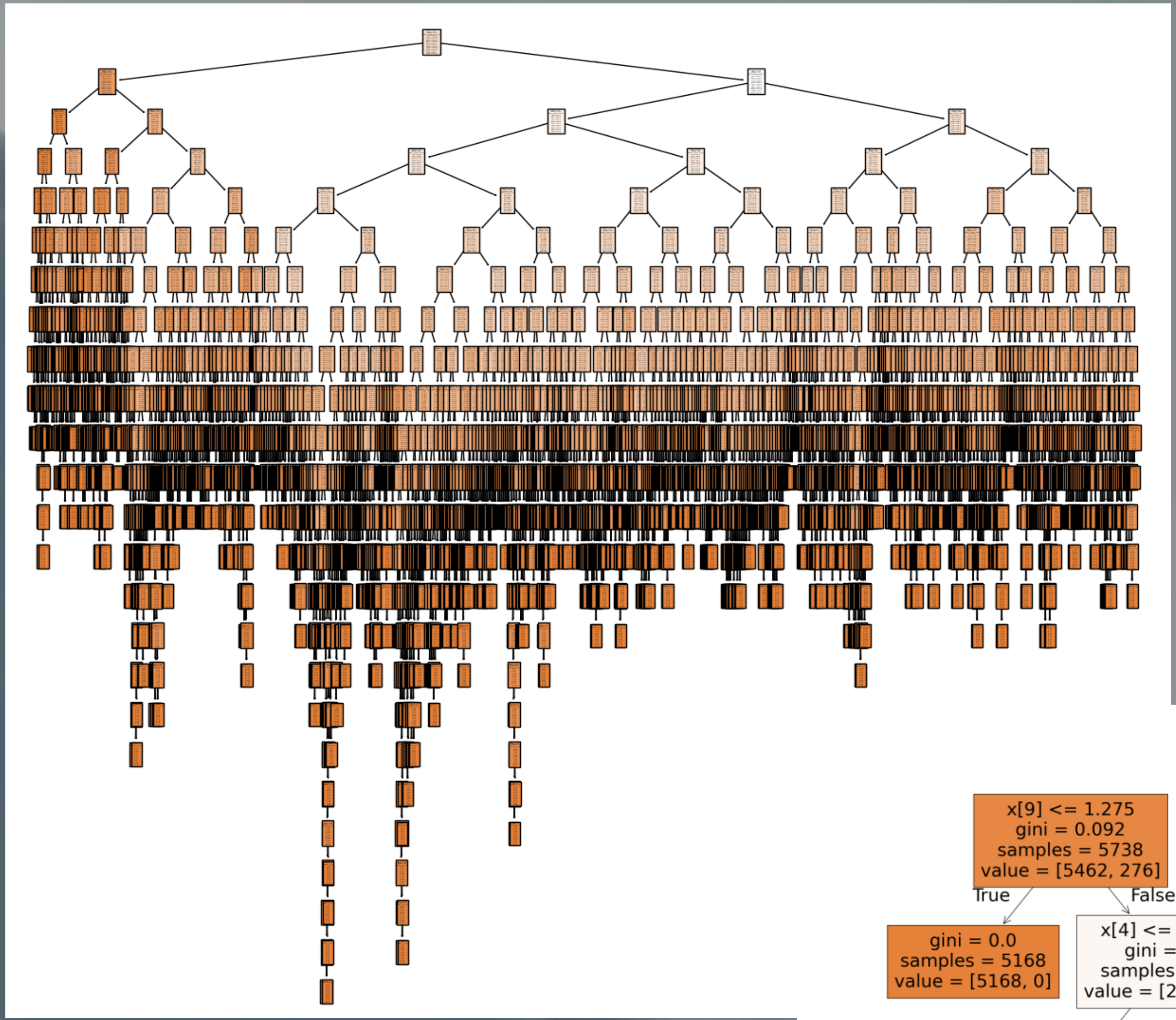
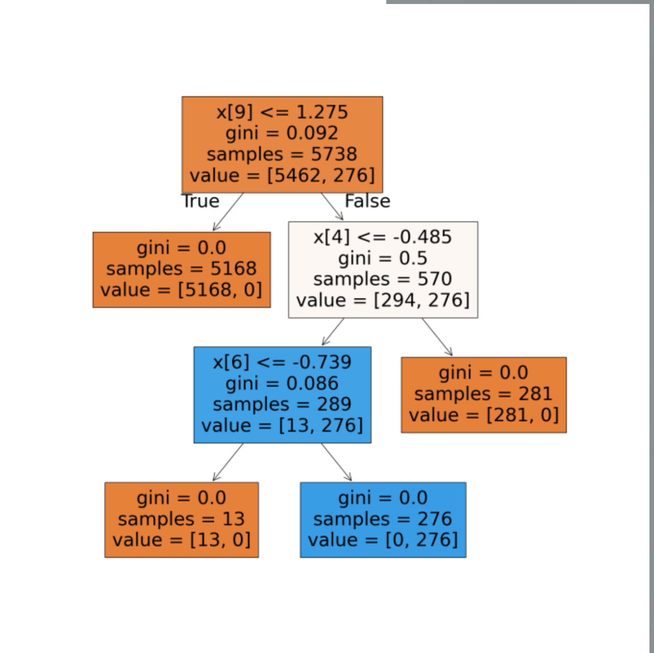
- K-Nearest Neighbor
- **Decision Tree**
- Artificial Neural Network





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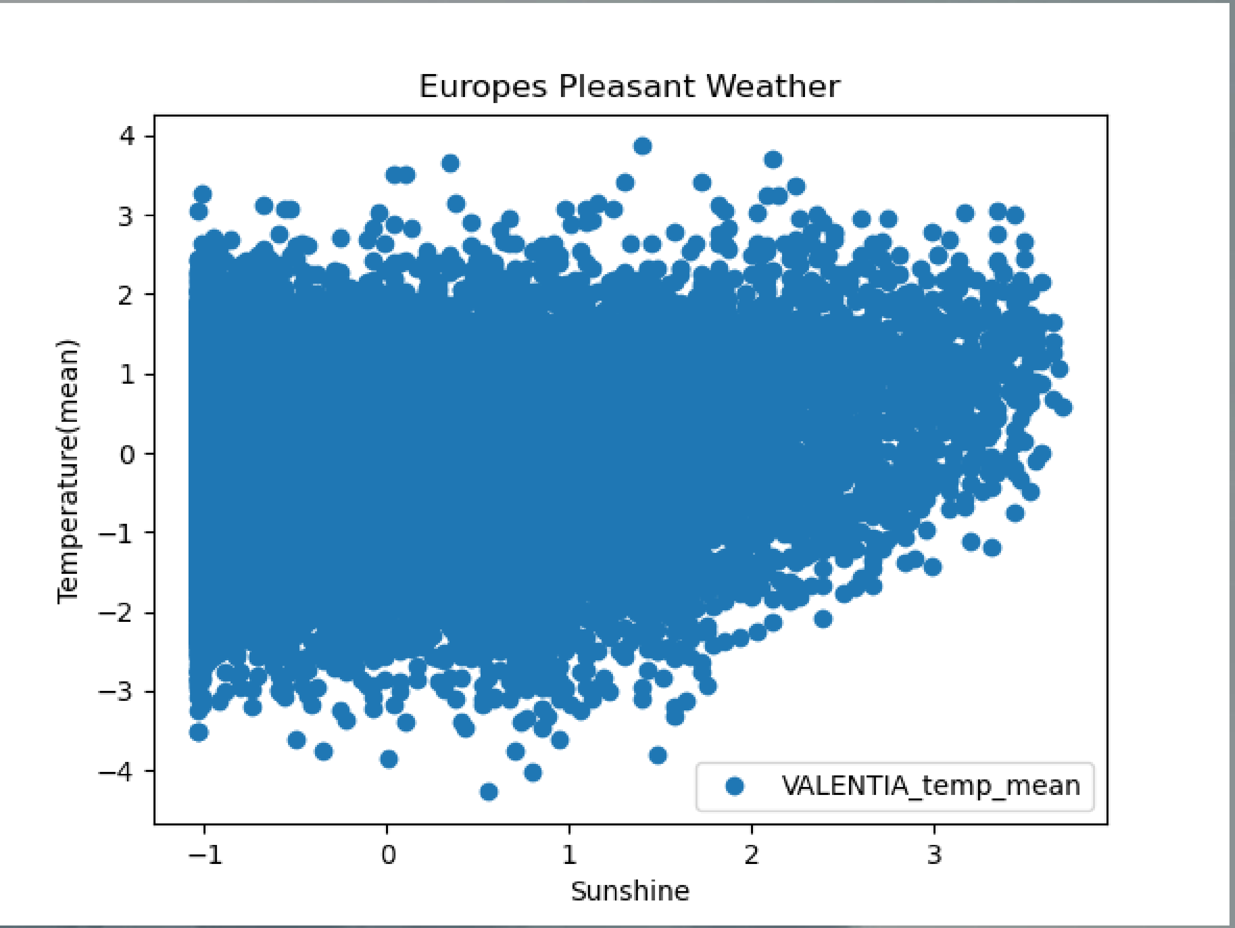

Supervised Learning Algorithms Best Fit

- K-Nearest Neighbor
- Decision Tree
- **Artificial Neural Network**

ANN Attempt 1

```
[68]: # Create the ANN
# hidden_layer_sizes has up to three layers, each with a number of nodes. So (5, 5) is two hidden layers with 5 nodes each,
# and (100, 50, 25) is three hidden layers with 100, 50, and 25 nodes.
mlp = MLPClassifier(hidden_layer_sizes=(5, 5), max_iter=500, tol=0.0001)

# Fit the data to the model
mlp.fit(X_train, y_train)

[68]: MLPClassifier
MLPClassifier(hidden_layer_sizes=(5, 5), max_iter=500)

[70]: y_pred = mlp.predict(X_train)
print(accuracy_score(y_pred, y_train))
y_pred_test = mlp.predict(X_test)
print(accuracy_score(y_pred_test, y_test))

0.4859981408319777
0.49477169745555943
```

ANN Attempt 2

```
[80]: # Create the ANN
# hidden_layer_sizes has up to three layers, each with a number of nodes. So (5, 5) is two hidden layers with 5 nodes each,
# and (100, 50, 25) is three hidden layers with 100, 50, and 25 nodes.
mlp2 = MLPClassifier(hidden_layer_sizes=(10, 5), max_iter=500, tol=0.0001)

# Fit the data to the model
mlp2.fit(X_train, y_train)

C:\Users\jmesi\anaconda3\Lib\site-packages\sklearn\neural_network\_multilayer_perceptron.py:690: ConvergenceWarning: Stochastic Optimizer: Maximum iterat
ions (500) reached and the optimization hasn't converged yet.
  warnings.warn(

[80]: MLPClassifier
MLPClassifier(hidden_layer_sizes=(10, 5), max_iter=500)

[83]: y_pred = mlp2.predict(X_train)
print(accuracy_score(y_pred, y_train))
y_pred_test = mlp2.predict(X_test)
print(accuracy_score(y_pred_test, y_test))

0.5223681152684174
0.5256186824677588
```

ANN Attempt 3

```
[90]: # Create the ANN
# hidden_layer_sizes has up to three layers, each with a number of nodes. So (5, 5) is two hidden layers with 5 nodes each,
# and (100, 50, 25) is three hidden layers with 100, 50, and 25 nodes.
mlp3 = MLPClassifier(hidden_layer_sizes=(20, 10, 10), max_iter=1000, tol=0.0001)

# Fit the data to the model
mlp3.fit(X_train, y_train)

[90]: MLPClassifier
MLPClassifier(hidden_layer_sizes=(20, 10, 10), max_iter=1000)

[92]: y_pred = mlp3.predict(X_train)
print(accuracy_score(y_pred, y_train))
y_pred_test = mlp3.predict(X_test)
print(accuracy_score(y_pred_test, y_test))

0.6706948640483383
0.6455210874869293
```

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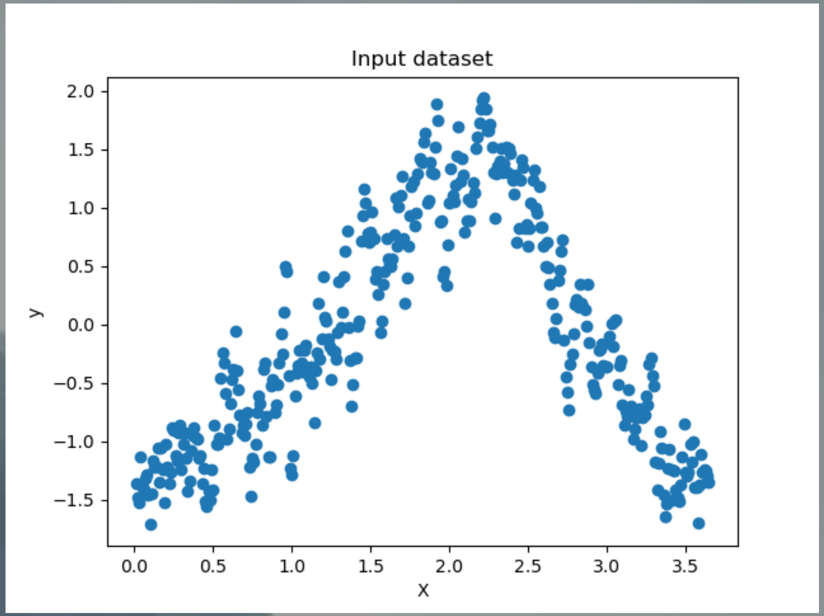
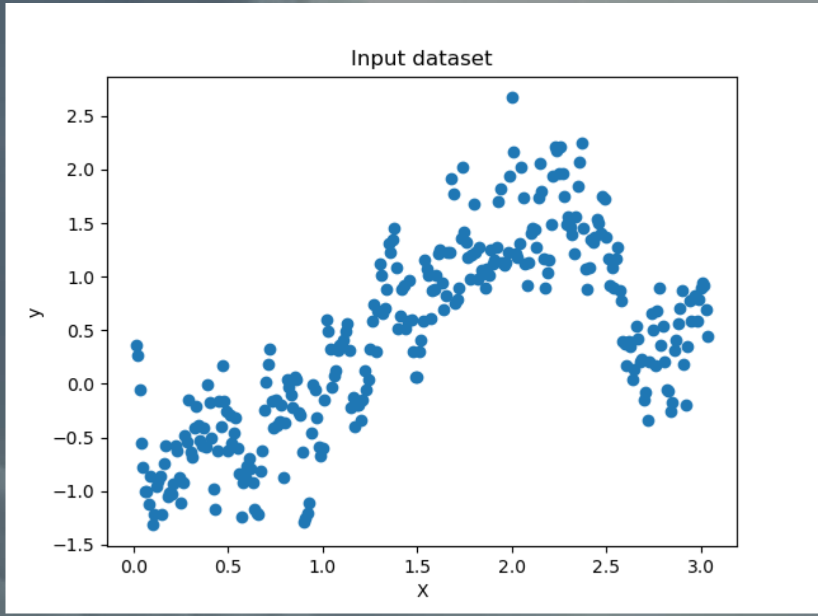
Summary

- **Hypothesis 1**: The climate hasn't changed much from 1973 until 2022.
  - The climate has begun to increase in temperature according to these scatterplots.



Madrid year 1973



Debilt year 2022

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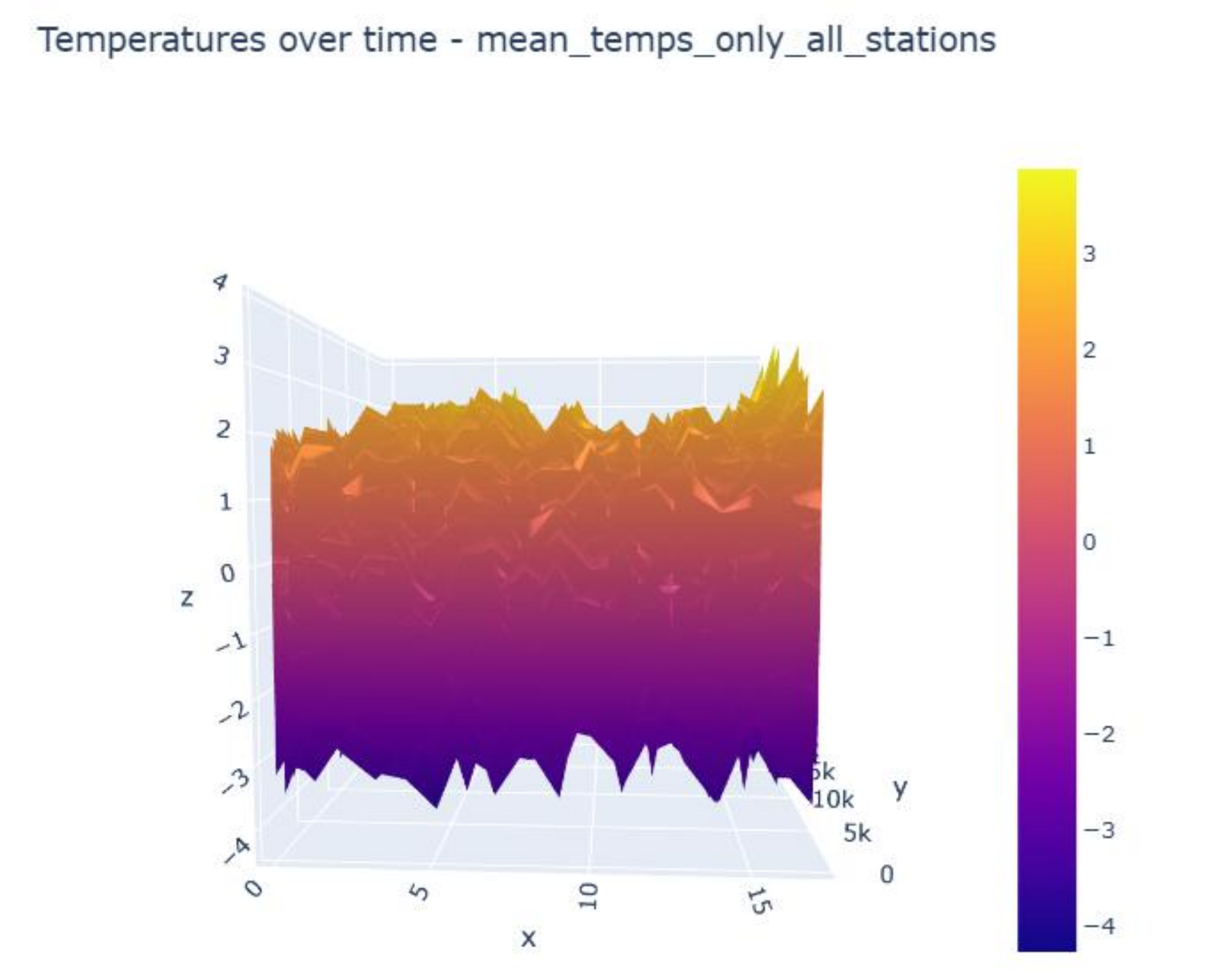
Summary

- **Hypothesis 2**: There was nicer weather about 50 years ago.
    - There is some temperature increase from years 1960-2022



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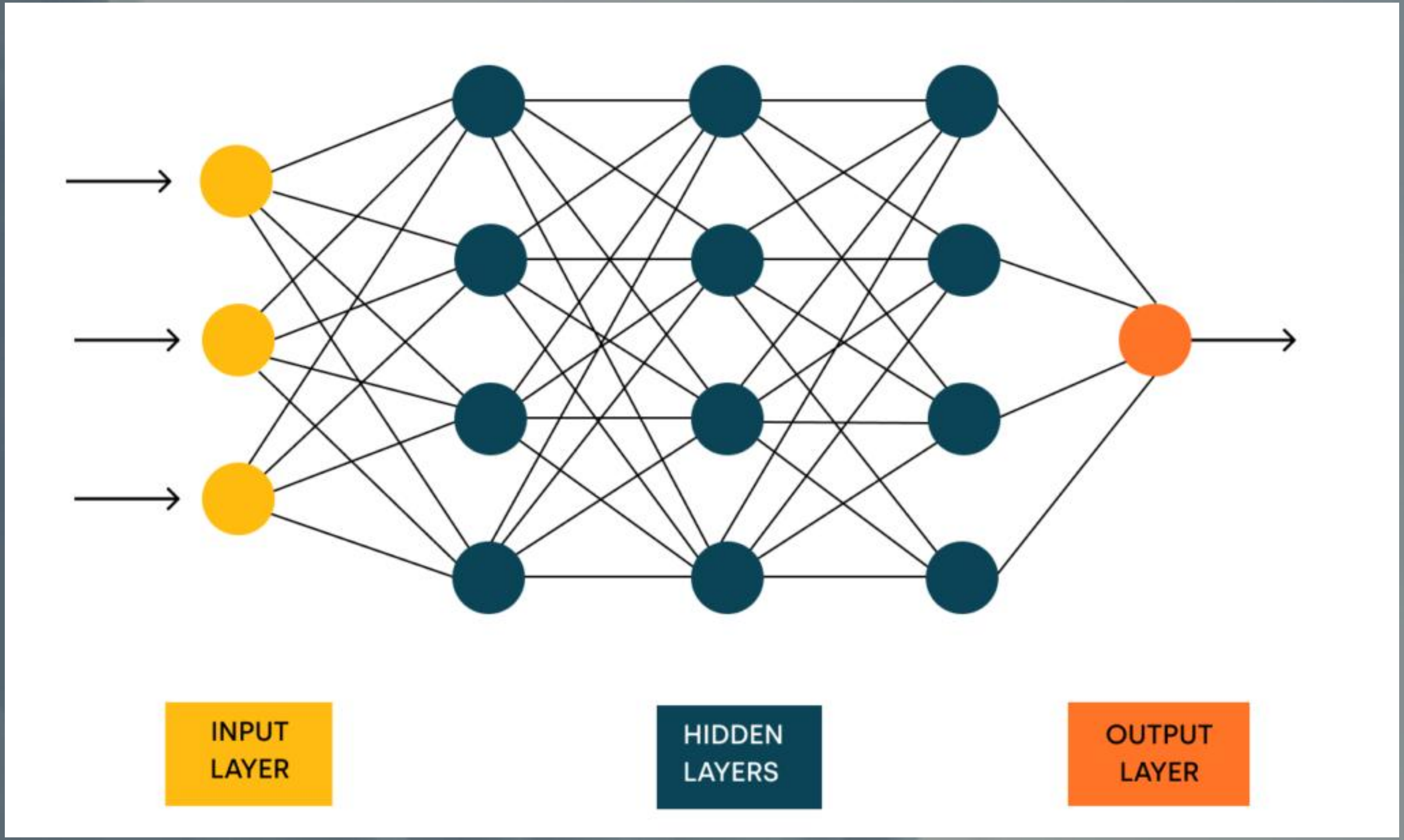
Summary

- **Hypothesis 3**: 70% of days in Europe aren't days with pleasant weather.
  - Majority of the time the average pleasant weather was 72%
- My chosen Machine Learning Algorithm is Artificial Neural Networks.
- Next steps would be to run a regression model on the larger dataset. Deeper dive into the data.



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Thank You!

Thank you for joining me on this machine learning journey.

I’m happy to take questions from the audience.

Check out my work for this project on Github.

Check out my LinkedIn

Explore my website outlining my projects.

Jordan Novelli

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